

Project Final

12/19/2025

Group 5

Topic: Redo experiments from Liu et al. (2023) that exploit sparsity to accelerate LLM inference

Outline

- Evaluate model sparsity on benchmarks with different levels of difficulty
- Evaluate sparsity of different model sizes on the same benchmark
- Evaluate sparsity of FFN and Attention in same model

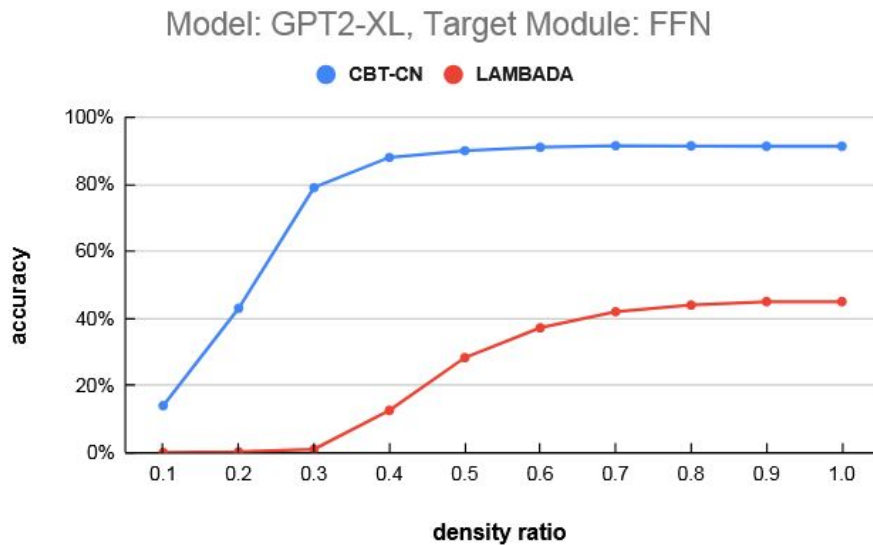
Model Sparsity on Tasks with Different Levels of Difficulty

- Task: image classification
 - CBT-CN: cloze test, choose 1 from 10
 - LAMNADA: word prediction, predict the last word (open-ended/whole vocabulary)
- Model: GPT2-XL with GELU-based FFN

Model Sparsity on Tasks with Different Levels of Difficulty

Experiment Results: **Harder tasks require more active parameters.**

- CBT-CN: activating 30% of parameters can achieve great accuracy.
- LAMBADA: activating 70% of parameters can achieve great accuracy.



Different Model Sizes on the Same Task

- Task:
 - Children's Book Test (CBT) – cloze test, choose 1 from 10
 - WikiText2 – Language Modeling, predict the next token (open-ended/whole vocabulary)
- Model: (we only evaluate the sparsity of the FFN layers)
 - GPT2 (56M parameters in FFN layers)
 - GPT2-XL (983M parameters in FFN layers)

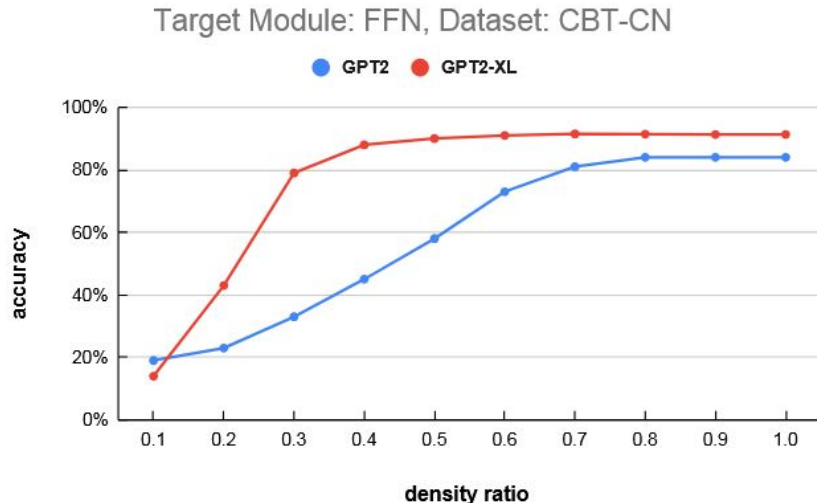
	gpt2	gpt2-xl
input dimension (d_{in})	768	1600
hidden dimension (h)	3072	6400
output dimension (d_{out})	768	1600

dimension of weight matrices of GPT-2

Different Model Sizes on the Same Task

Experiment Results: **Both model have similar size of hidden layers to achieve 80% accuracy.**

- GPT2: activating 70% (~2150) of FFN hidden layers achieve 80% accuracy.
- GPT2-XL: activating 30% (~1920) of FFN hidden layers achieve 80% accuracy.



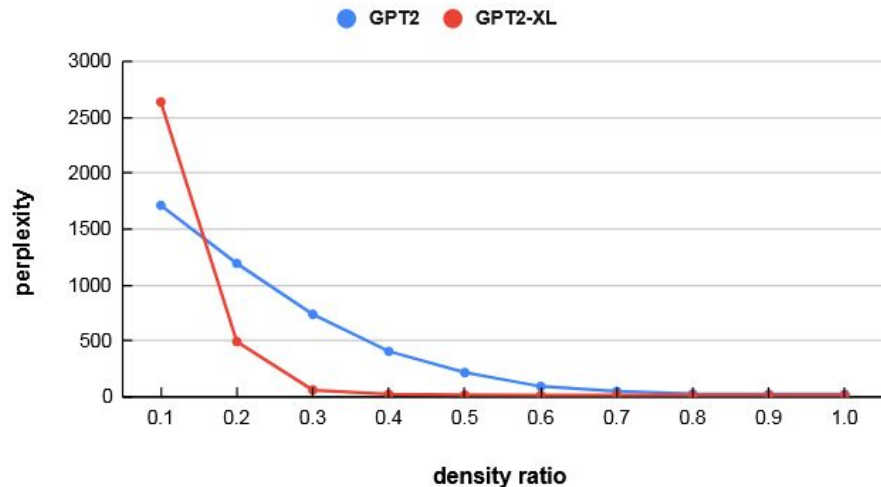
- GPT-2 (Small): $3072 \times 0.7 \approx 2150$
- GPT-2 XL (Large): $6400 \times 0.3 = 1920$

Different Model Sizes on the Same Task

Experiment Results: **Both model have similar size of hidden layers to achieve 500 perplexity.**

- GPT2: activating 40% (~1228) of FFN hidden layers achieve 80% accuracy.
- GPT2-XL: activating 20% (~1280) of FFN hidden layers achieve 80% accuracy.

Target Module: FFN, Dataset: WikiText2



- GPT-2 (Small): $3072 \times 0.4 \approx 1228$
- GPT-2 XL (Large): $6400 \times 0.2 = 1280$

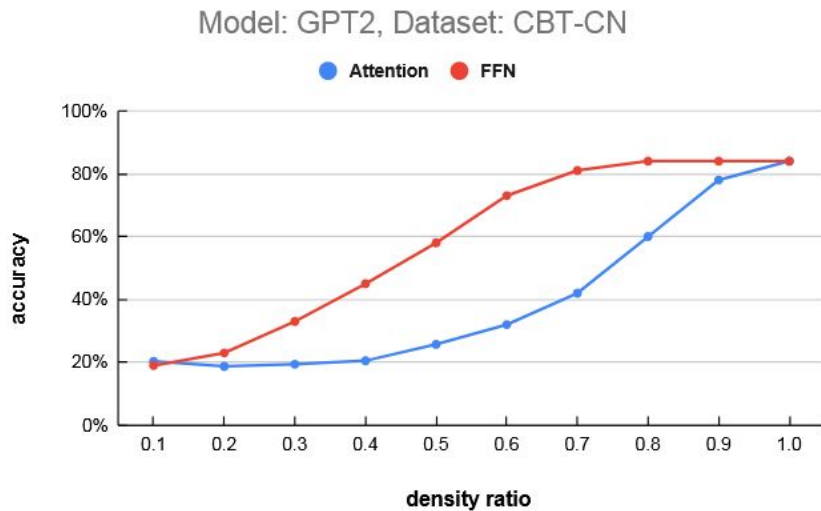
Attention vs. Feed-Forward Sparsity

- Task:
 - Children's Book Test (CBT) – cloze test, choose 1 from 10
 - MGSM: Math reasoning, solve multi-step arithmetic word problems
- Model:
 - GPT2 (Uses standard Multi-Head Attention and basic GELU-based FFN.)
 - GPT2-XL (Identical architecture to GPT-2)
 - Llama3-8B (Adopts Grouped-Query Attention and SiLU-based FFN)

Attention vs. Feed-Forward Sparsity

Experiment Results: **FFN parameters can be sparsified than Attention weights in GPT2 model**

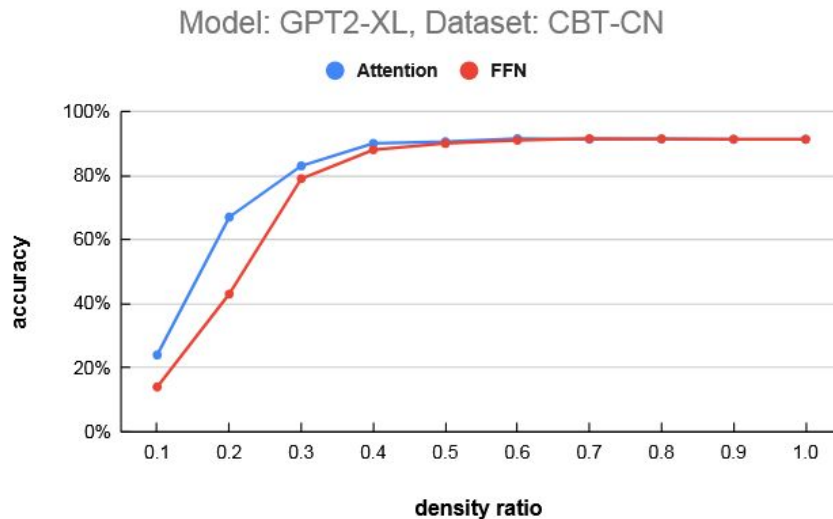
- FFN: achieve 80% accuracy when activate 70% of FFN parameters.
- Attention: achieve 40% accuracy when activate 70% of Attention parameters.



Attention vs. Feed-Forward Sparsity

Experiment Results: **Attention not only withstands aggressive compression but surpasses the FFN module's robustness in GPT2-XL model**

- FFN: achieve 42% accuracy when activate 20% of FFN parameters.
- Attention: achieve 65% accuracy when activate 20% of Attention parameters.



Attention vs. Feed-Forward Sparsity

Experiment Results: **FFN weights can be more aggressively sparsified than Attention weights in Llama3-8B model**

- FFN: activating 30% of FFN parameters achieve 55% accuracy.
- Attention: activating 80% of Attention parameters achieve 55% accuracy.

Model: Llama3-8B, Dataset: MGSM

